

ANTHONY CATALINA

Full-Stack Engineer · Embedded + Frontend · IoT & Device Platforms | Fort Collins, CO · (609) 651-0935 ·
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SUMMARY

Full-stack engineer with ~5 years of experience building production web applications and device-connected software. Unique blend of embedded systems depth and modern frontend expertise; comfortable owning end-to-end features from firmware integration through React UI. Strong track record delivering real-time IoT dashboards, design systems, and REST APIs for enterprise clients.

TECHNICAL SKILLS

- **Frontend:** React, TypeScript, JavaScript, Lit, HTML/CSS/SCSS, Bootstrap, Atomic UI, MVC
- **Backend & APIs:** Python, Flask, REST API design, SQL
- **Embedded / IoT:** Embedded C/C++, MQTT, Bluetooth/BLE, firmware workflows, device data pipelines
- **Cloud & DevOps:** AWS (S3, GitHub Actions CI/CD), Git, Jenkins, Bitbucket, Jest, ESLint

PROFESSIONAL EXPERIENCE

Embedded Software Engineer · Hach Company

Jan 2026 – Present

Contract Firmware Engineer · Hach Company / Actalent

Apr 2025 – Jan 2026

Loveland, CO · Hybrid

- Combining embedded systems development with frontend architecture for real-time device management and firmware workflow interfaces; bridging hardware data to UI end-to-end.
- Maintained a large-scale TypeScript/Lit monorepo across multiple water analysis product lines; integrated live device data via MQTT and HTTP for accurate, low-latency UI updates.
- Implemented custom ESLint rules, Jest unit test suites, and managed UI submodules across concurrent projects; enforcing Atomic UI and MVC architecture team-wide.

TypeScript · Lit · React · MQTT · Embedded C/C++ · Jest · ESLint · Jenkins · Bitbucket

Software Engineer · Philadelphia Scientific, LLC

Sept 2022 – Apr 2025

Montgomeryville, PA

- Led full-stack development of production web apps for Fortune 500 clients (Target, TJX, Uline); React/TypeScript frontends, Python/Flask APIs, and SQL backends designed and deployed independently.
- Architected a real-time operations dashboard with interactive charts and live data feeds; deployed to AWS S3 with zero-downtime CI/CD via GitHub Actions.
- Built a touch-optimized React app for on-device battery maintenance, rolled out across major distribution centers via SureMDM MDM; provided ongoing production support and customer troubleshooting.

React · TypeScript · Python · Flask · SQL · AWS · Bootstrap · GitHub Actions · SureMDM

Software Engineer · Regulus Group, LLC

Jun 2021 – Sept 2022

Atlantic City, NJ

- Upgraded hardware/software capabilities for FAA Surveillance Branch data processing; rebuilt data-input and analytical-output UIs for government operators.
- Developed real-time and historical system performance dashboards; optimized SQL queries for data-intensive workloads.

JavaScript · HTML/CSS · SQL · Dashboard development

NOTABLE PROJECTS

Water Analysis UI — Hach TypeScript · Lit · React · MQTT · Jest

- Real-time instrument UI serving engineers, lab techs, and clients across multiple product lines; modular Atomic UI components reused across product lines.

Battery Manager Operations Display — Philadelphia Scientific React · TypeScript · Python · SQL · AWS

- Real-time ops dashboard for enterprise distribution centers; live data visualization + automated AWS/GitHub Actions deployment pipeline.

Battery Manager Touch App — Philadelphia Scientific React · TypeScript · Python · Flask · SQL

- Touch-optimized on-device web app with full REST API backend; deployed across Target, TJX, and Uline via SureMDM.

EDUCATION

B.S. Computer Science · Rowan University · Glassboro, NJ

Dec 2020